# Mohammad Adil Abdulrashid

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## Education

**Bachelor of Science in Electrical Engineering** 2023 – 2028 (expected)
*Qatar University, Doha, Qatar*

- **GPA:** 3.50 / 4.00

## Technical Skills

**Embedded and hardware:** ESP32, Arduino, embedded C/C++, sensor interfacing, motor control, NI ELVIS, 3D printing
**Programming:** Python (FastAPI, TensorFlow), VHDL, GDScript
**Design and simulation:** SolidWorks, OpenSCAD, Blender, MATLAB, Multisim, Digital (logic simulator)
**Tools:** Git, Microsoft Office
**Languages:** English, Urdu, Hindi

## Honours and Awards

**Best Research Project Award, Track 3** 2026
*Summer Research Internship Program, Qatar University (Sensebridge project)*

**Outstanding Performance Scholarship**
*Qatar University, awarded to the top 50 students university-wide*

**Dean's List and Vice President's List**
*Qatar University*

## Research Experience

**Sensebridge: AI-Assisted Speech Therapy** Summer 2026
*Summer Research Intern, Qatar University, Doha*

- Developed a lightweight speech-sound classifier to support children's speech therapy, recognising the /s/, /m/ and /p/ phonemes; trained a convolutional neural network on audio spectrograms using transfer learning in TensorFlow, sized for future on-device use in a mobile app and custom circuit.
- Built the training dataset from recordings of around 10 paediatric participants and reached 70% (±2%) classification accuracy in testing with children.
- Worked in a multidisciplinary team with a speech therapist, computer science students and an electrical and environmental engineering mentor; presented the results in a final report and oral presentation.

**Non-Invasive Continuous Blood Glucose Monitoring** 2026 – present
*Undergraduate Researcher with Dr. Muhammad E. H. Chowdhury, Qatar University, Doha*

- Contributing to hardware prototyping and optical sensor integration for photoplethysmography (PPG) signal acquisition.
- Wrote a literature review covering the evolution of analysis methods, multimodal sensor-fusion pipelines, the underlying hardware physics, and clinical and regulatory benchmarks for non-invasive glucose estimation.

**Smart Corrosion-Sensing Coatings** 2026 – present
*Undergraduate Research Experience Program (UREP), Qatar University, Doha*

- Synthesised $CaCO_3$ microspheres by precipitation to encapsulate a corrosion indicator (phenolphthalein) and a corrosion inhibitor (5-phenyl-1H-tetrazole).
- Formulated and cast polyurethane coatings containing the microspheres, then ran electrochemical and visual corrosion tests on carbon-steel samples to measure sensing and inhibition performance.
- Iterated the design with graduate supervisors, building skills in experimental analysis, technical documentation and research methods.

## Engineering Projects

**DustMap: Dust Inspection Platform for Industrial Sites** 2026
*Team project for Qatar's Ministry of Environment and Climate Change*

- Co-designed the portable monitoring units with the team: an ESP32 and particulate-matter sensor in a 3D printed enclosure, reporting readings to the monitoring platform.

- Developed the team's initial prototype into a complete inspection platform for quarries, crushers and concrete plants, covering site modelling, field inspections, hotspot tracking across repeat visits and printable reports (Python, FastAPI, SQLite).
- Designed a two-monitor measurement method: a handheld unit surveys approved points while an upwind reference unit records background dust, so each reading reflects the site's own contribution.
- Built a Gaussian-process dust map that recommends the next point to measure; across 12 simulated sites it found 71% of hotspots versus 55% for a fixed route and 50% for random sampling, with 91% flag precision.
- Developed a 2D advection–diffusion dust simulator (PM2.5 and PM10, five source types, building obstruction) to test the platform before sensor hardware was available, and validated it against the Gaussian plume equation to within 1% (centreline) and 2% (plume width).

**Sim Racing Steering Wheel and Pedal** 2026

*Personal project*

- Designed and 3D printed a steering wheel and foot pedal that pair with a PC as an ESP32-based Bluetooth gamepad.
- Iterated through three sensing methods, from a capacitive foil slider to pencil-graphite resistive tracks to geared potentiometers in the final design, after the graphite track proved too noisy and unreliable.
- Geared each potentiometer to use more of its range: 2:1 on the steering (±45° at the wheel turns the potentiometer ±90°) and 6:1 on the pedal, giving about 2,400 and 1,000 usable ADC steps.
- Modelled all parts parametrically in OpenSCAD, refining fits across revisions so every part prints without supports.

**AIChE ChemE Car Competition** 2026 – present

*American Institute of Chemical Engineers (AIChE) and Qatar University; electrical engineer on the QU team*

- Designed the car's control electronics, using a microcontroller to drive the motor and to time the chemical reaction that determines the stopping distance.
- Implemented dynamic braking by shorting the motor terminals through the motor driver at the stop signal, eliminating coast-down and improving stopping accuracy.

**8-Bit Computer Designed from Logic Gates** 2026

*Gate-level digital logic simulator*

- Designed a complete 8-bit computer bottom-up, building every component from NAND gates: logic gates, multiplexers and decoders, adders, and memory from latches and flip-flops.
- Built the datapath: an 8-bit arithmetic logic unit (ALU) supporting addition, subtraction and four logic operations, six general-purpose registers with input and output ports, and a program counter.
- Designed the control logic: an instruction decoder in which the top two bits of each instruction select an immediate load, ALU operation, register copy or conditional jump, and a condition unit that evaluates eight jump conditions on the result.
- Wrote and debugged assembly programs on the finished machine, verified against the simulator's test cases.

**Continuous Syringe Rotation Device** 2026

*Built for a graduate research lab, Qatar University*

- Identified that multi-component samples were separating inside the syringe during a graduate researcher's experiments, with no off-the-shelf fix available, and designed one.
- Designed the mount in SolidWorks, 3D printed the housing and custom GT2 pulley clamps, and assembled a NEMA 17 stepper drive with an Arduino Uno and GT2 belt, at a fraction of the cost of commercial alternatives.
- Validated the device in operation; it maintained fluid homogeneity throughout continuous infusion.

**Automated Irrigation and Environmental Monitoring System** 2026

*ESP32 embedded system*

- Built an ESP32 data logger reading a DS18B20 temperature sensor, a DHT22 humidity sensor and a capacitive soil-moisture sensor, recording timed measurements to an SD card.
- Conducted a controlled growth study to determine the optimal soil-moisture threshold, then implemented closed-loop control of a servo-driven valve based on live sensor readings.

**Moribund: Isometric Action RPG** 2026 – present

*Solo project, Godot 4 (GDScript)*

- Developing a gothic isometric action RPG independently, with all scenes, levels and interfaces generated in code.
- Built a dynamic lighting system with line-of-sight torch shadows, fog, weather effects and custom post-processing shaders, and an asset pipeline that renders 3D models in Blender into 16-direction sprites with normal maps.
- Implemented enemy AI for 27 monster types and six boss encounters, two player classes with talent trees, and a branching quest and dialogue system.

## Training and Workshops

**Nokia Telecommunications Technology Workshop** 2024

*Delivered by Nokia*

- Covered mobile network evolution from 2G to 5G (GSM, WCDMA, LTE), 5G use cases, network slicing and 6G research.